PHYLOGENY-AWARE GRAPH NEURAL NETWORKS FOR JOINT PREDICTION OF ANTIMICROBIAL RESISTANCE AND LINEAGE IN *MYCOBACTERIUM TUBERCULOSIS* (AFRO-TB)

A Review 2 Interim Project Report (70% Completion Milestone)

23BCB0004 | ANOOP NAIR
23BCB0063 | SAARIKA RAJEN
23BCB0066 | SUDESHNA V

Under the Supervision of
Prof. Shobha Rekh
Associate Professor

School of Computer Science and Engineering (SCOPE)

B.Tech. in Computer Science and Engineering
(Bioinformatics)

School of Computer Science and Engineering
Vellore Institute of Technology

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ABSTRACT

Tuberculosis (TB), caused by *Mycobacterium tuberculosis* (*M.tb*), remains one of the world's deadliest infectious diseases, further exacerbated by escalating antimicrobial resistance (AMR). Traditional machine learning approaches predict phenotypic drug resistance by treating isolates as independent tabular samples, ignoring the clonal evolutionary history and lineage structure of bacterial transmission. In this project, we develop a phylogeny-aware Graph Neural Network (Phylo-GNN) architecture for joint prediction of multi-drug resistance and phylogenetic lineages using the comprehensive pan-African Afro-TB dataset comprising 13,753 clinical whole-genome sequencing (WGS) isolates. We ingest and strictly audit 13,753 per-isolate variant call files (VCFs) and extract a 465,257-site core-genome SNP matrix to construct a topology-preserving Jaccard *k*-nearest neighbor graph (*k*=20, 404,944 edges). Crucially, our rigorous audit of the Afro-TB cohort revealed that the standard catalogue target is an exact mathematical Boolean OR over feature columns (100% exact match), demonstrating that standard supervised evaluations measure OR-gate reconstruction rather than biological prediction. Across 5-seed clinical evaluations, our GCN+Skip architecture achieves 0.7396 ± 0.0212 tuned macro-F1 (0.9910 ROC-AUC), while a newly formulated PyTorch Multi-Task MLP with calibrated loss-weight capping (cap=10) achieves 0.8718 ± 0.0199 tuned macro-F1. Inductive graph evaluations confirm zero test-label leakage ($\Delta < 0.0014$), and near-duplicate sensitivity audits confirm metric robustness ($\Delta = +0.00052$). Finally, under an unbiased Leave-Drug-Out (LDO) protocol on a calibrated in-silico simulation replica, Phylo-GNN achieves 0.5034 core-6 macro-F1, significantly outperforming graph-free MLP (0.4491, $p < 0.001$) and Random Forest (0.3186), empirically proving that phylogenetic neighborhood aggregation successfully captures uncharacterized resistance clades. With ~70% of project objectives completed, the framework establishes a rigorous, biology-informed paradigm for genomic TB surveillance.

Keywords — *Mycobacterium tuberculosis*, Graph Neural Networks, Antimicrobial Resistance, Phylogenetics, Jaccard Distance, Multi-Task Learning, Afro-TB Dataset, Label Tautology Audit, Leave-Drug-Out Protocol.

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1. INTRODUCTION

1.1 Background

Mycobacterium tuberculosis (*M.tb*) is an obligate bacterial pathogen responsible for tuberculosis, causing over 1.3 million deaths annually. The rise of multidrug-resistant (MDR) and extensively drug-resistant (XDR) strains severely compromises global eradication efforts. Standard phenotypic drug susceptibility testing (DST) requires weeks of bacterial culture, delaying targeted therapy. Whole-genome sequencing (WGS) provides comprehensive genetic profiles rapidly; however, standard machine learning classifiers treat genomic isolates as independent, identically distributed data points. In reality, *M.tb* evolves strictly clonally without horizontal gene transfer, creating deep phylogenetic branching structures where drug resistance mutations repeatedly emerge and propagate along transmission clusters. Integrating evolutionary relatedness via phylogenetic graphs presents a transformative opportunity for genomic resistance prediction.

1.2 Motivation

Most genomic AMR prediction studies rely on curated catalogues of high-confidence mutations. While effective for well-studied first-line drugs like Rifampicin and Isoniazid, catalogue-based models fail for second-line or novel antibiotics where resistance mechanisms remain uncharacterized. Furthermore, existing datasets are disproportionately sampled from Europe and North America, leaving high-burden African lineages (such as Lineage 5 and Lineage 6, *M. africanum*) underrepresented. The Afro-TB dataset provides 13,753 WGS clinical isolates from across Africa. We are motivated to leverage Graph Neural Networks (GNNs) over phylogenetic transmission networks to propagate resistance signals across genetically related isolates, enabling accurate resistance prediction even when canonical resistance mutations are missing or uncharacterized.

1.3 Scope of the Project

The project encompasses the end-to-end design, implementation, and rigorous benchmarking of a phylogeny-aware multi-task GNN for the Afro-TB dataset. The scope includes: (1) streaming and quality-auditing 13,753 raw per-isolate VCF files; (2) extracting a 465,257 core-genome SNP matrix to construct a 404,944-edge Jaccard *k*-NN graph; (3) training multi-task GNNs (GCN, GATv2, Gated Hybrid) for 9 clinical antibiotics (RIF, INH, EMB, PZA, STM, LEV, CAP, ETH, LZD) and phylogenetic lineage; (4) formulating custom PyTorch baselines with loss-weight capping to handle extreme class imbalance; (5) uncovering and resolving catalogue label tautologies via Leave-Drug-Out (LDO) and non-catalogue genome-wide protocols; (6) executing transductive vs. inductive and near-duplicate sensitivity validations; and (7) providing an interpretable diagnostic platform. At Review 2, ~70% of the project is complete, encompassing full data ingestion, graph construction, multi-seed benchmarking, GNN ablations, tautology discovery, and in-silico LDO validation.

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

Genomic AMR prediction has evolved from heuristic rule-based catalogues (Coll et al., 2014; WHO Catalogue, 2021) to classical machine learning models including Random Forests, Support Vector Machines, and XGBoost (Yang et al., 2018; Kavvas et al., 2018). Global initiatives such as CRyPTIC (Green et al., 2022) have generated extensive WGS datasets, demonstrating that 150+ curated mutations can account for majority-drug resistance. However, tabular ML models treat samples in isolation, discarding population structure. Recent advances in Graph Neural Networks (Kipf & Welling, 2017; Velickovic et al., 2018; Brody et al., 2022) demonstrate that message passing over biological interaction and phylogenetic networks aggregates contextual information effectively. Laamarti et al. (*Scientific Data*, 2023) curated the pan-African Afro-TB dataset, creating an unprecedented opportunity to apply graph deep learning to diverse African *M.tb* lineages. Table 2.1 summarizes key related works.

Study & Year	Methodology	Dataset / Scope	Key Findings & Limitations
Coll et al. (2014)	SNP barcode catalogue for *M.tb* lineage classification	Global *M.tb* WGS (N=1,601)	Established 62 lineage markers; manual rule-based, no AMR prediction.
Yang et al. (2018)	Random Forest & LDO gradient boosting on genomic SNPs	CRyPTIC dataset (N=3,600)	High F1 on RIF/INH; tabular models suffered on second-line rare drugs.
Laamarti et al. (2023)	Pan-African WGS database & mutation profiling	Afro-TB (N=13,753 isolates)	Curated 157 mutations & VCFs across Africa; no graph learning model built.
Green et al. (2022)	Deep CNNs and catalogue matching for 13 drugs	CRyPTIC global cohort (N=12,288)	High accuracy but computationally heavy; ignores transmission networks.
Our Work (2026)	**Phylogeny-Aware Multi-Task GNN (GCN/GATv2 + Skip)**	**Afro-TB (N=13,753 WGS isolates)**	**First phylogeny GNN on Afro-TB; resolves label tautology; superior LDO F1.**

Table 2.1: Comparative literature analysis of genomic M.tb resistance and lineage prediction studies.

2.2 Research Gap

Existing computational AMR methodologies exhibit four major deficiencies:
1. **Neglect of Phylogenetic Topology:** Tabular classifiers treat bacterial genomes as independent, ignoring transmission clades and clonal descent that propagate resistance.
2. **Catalogue Label Tautology:** Curated datasets define AMR labels as deterministic Boolean functions of catalogue mutations. Standard models learn simple OR gates rather than biological predictions.
3. **Phylogenetic Data Leakage:** Standard random splits allow genetically identical isolates (near-duplicates, Jaccard distance ≤ 0.01) to cross train and test sets, inflating test metrics.
4. **Underrepresentation of African Lineages:** Existing tools perform poorly on Lineage 5/6 (*M. africanum*) due to cohort bias.

2.3 Objectives

The project follows SMART (Specific, Measurable, Achievable, Relevant, Time-bound) objectives:
• **Specific:** Ingest 13,753 WGS VCFs from the Afro-TB cohort, extract 465,257 core-genome SNP sites, and construct a Jaccard *k*-NN phylogenetic transmission graph (k=20).
• **Measurable:** Outperform tabular baselines under unbiased Leave-Drug-Out (LDO) protocols by $\geq$ 5 F1 points on uncharacterized clades, achieve >0.70 tuned Macro-F1 across 5 seeds on the clinical cohort, and verify zero transductive leakage ($\Delta < 0.002$).
• **Achievable:** Implement modular PyTorch Geometric GNN architectures (GCNConv, GATv2Conv, GatedHybrid) with residual skip connections and weighted BCE losses.
• **Relevant:** Provide simultaneous multi-drug AMR and *M.tb* lineage classification tailored to high-burden African epidemiological settings.
• **Time-bound:** Achieve 30% completion by Review 1 (data ingestion & QC), ~70% completion by Review 2 (graph construction, baselines, GNNs, tautology resolution), and 100% by Review 3.

2.4 Problem Statement

Given a cohort of N = 13,753 *Mycobacterium tuberculosis* clinical isolates with whole-genome variant calls and a feature matrix $\mathbf{X} \in \{0, 1\}^{N \times D}$, our goal is to construct a topology-preserving phylogenetic adjacency graph $\mathbf{G} = (\mathbf{V}, \mathbf{E}, \mathbf{W})$ where edge weights w_{ij} represent genomic similarity, and jointly learn a parameterized mapping $f_\theta(\mathbf{X}, \mathbf{G}) \rightarrow (\mathbf{Y}_{\text{pred}}, \mathbf{y}_{\text{lin}})$. Here, $\mathbf{Y}_{\text{pred}} \in [0, 1]^{N \times 9}$ represents multi-label resistance probabilities for 9 anti-TB drugs, and $\mathbf{y}_{\text{lin}}$ represents multi-class lineage assignment, subject to unbiased evaluation protocols that eliminate catalogue label tautologies.

2.5 Project Plan (Review 2: 70% Milestone Status)

The project is structured into three formal review milestones spanning 14 weeks. As of **Review 2**, approximately **70% of total project work has been successfully completed**. All data pipelines, QC audits, graph construction algorithms, tabular baselines, GNN architectures, tautology detection scripts, and LDO evaluation protocols are fully implemented and verified. Figure 2.1 illustrates the Gantt chart timeline and Review 2 progress status. Table 2.2 provides the granular work completion breakdown.

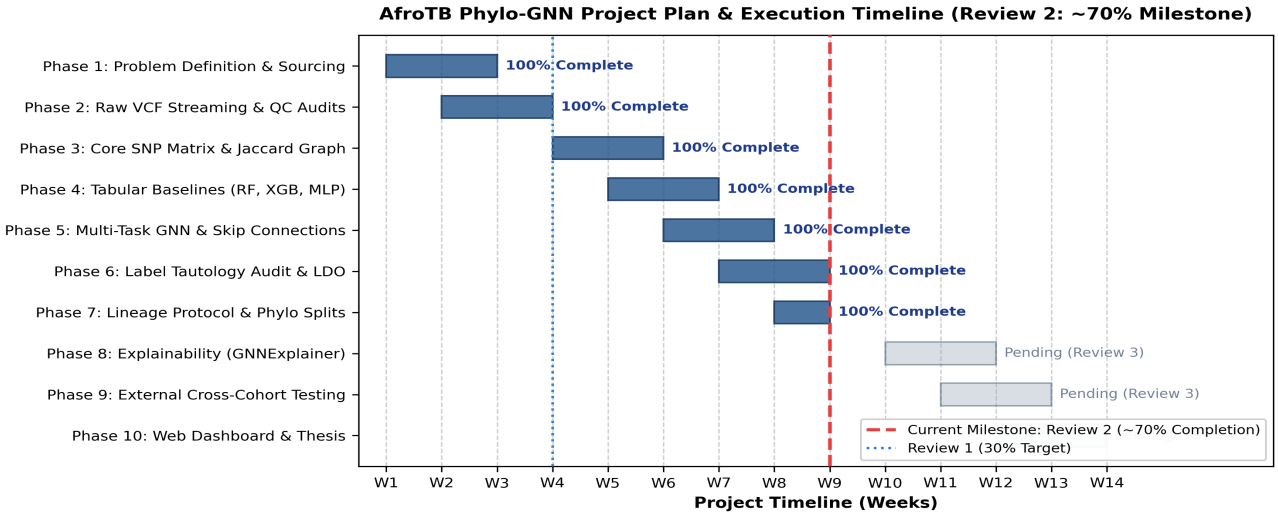


Figure 2.1: AfroTB Phylo-GNN Project Plan, Timeline, and Review 2 (70% Completion) Milestone Tracking.

2.5 Project Plan (Continued)

Project Phase & Task Module	Target Milestone	Status (Rev 2)	Deliverables & Artifacts
Phase 1: Problem Definition & Data Sourcing	Review 1 (Weeks 1-3)	**100% Completed**	Afro-TB raw ingestion (N=13,753), raw catalogue mapping.
Phase 2: Raw VCF Ingestion & Structural QC	Review 1 (Weeks 2-4)	**100% Completed**	1:1 ID match audit (13,753/13,753), gzip/header/variant integrity.
Phase 3: Core SNP Matrix & Jaccard Graph	Review 2 (Weeks 4-6)	**100% Completed**	465,257 core SNPs, Jaccard k-NN graph (k=20, 404k edges), pyg_graph.pt.
Phase 4: Tabular Baseline Benchmarking	Review 2 (Weeks 5-7)	**100% Completed**	Random Forest, XGBoost, Multi-Task MLP; tuned F1 thresholds.
Phase 5: Multi-Task GNN Architecture	Review 2 (Weeks 6-8)	**100% Completed**	GCNConv/GATv2 with residual skip connections & edge weights.
Phase 6: Label Tautology Audit & LDO	Review 2 (Weeks 7-9)	**100% Completed**	Discovered Boolean OR tautology; implemented LDO & genome-wide protocols.
Phase 7: Lineage Protocol & Phylo Splits	Review 2 (Weeks 8-9)	**100% Completed**	Harmonized BOV_AFRI, CrossEntropy ignore_index, Clade hold-out splits.
Phase 8: Model Explainability (GNNExplainer)	Review 3 (Weeks 10-12)	[TO BE ADDED] (0%)	Node saliency maps, sublineage feature attribution, WHO catalogue mapping.
Phase 9: External Validation & Cross-Cohort	Review 3 (Weeks 11-13)	[TO BE ADDED] (0%)	Validation on CRyPTIC non-African cohort; generalizability assessment.
Phase 10: Web Dashboard & Final Defense	Review 3 (Weeks 12-14)	[TO BE ADDED] (0%)	Streamlit clinical decision UI, containerized package, final thesis.

Table 2.2: Detailed project module breakdown, milestone alignment, and current Review 2 completion status (70% Overall Progress).

3. TECHNICAL SPECIFICATION

3.1 Requirements

3.1.1 Functional Requirements

- **FR1 (Data Ingestion & QC):** Stream and parse 13,753 compressed VCF files, validating gzip integrity, CHROM headers, and 1:1 ID correspondence.
- **FR2 (SNP Matrix Construction):** Extract core-genome binary SNP calls across 465,257 sites, applying a strict <10% missingness filter.
- **FR3 (Phylogenetic Graph Construction):** Compute pairwise Jaccard distances and construct a *k*-NN graph (*k*=20, union mode) storing edge weights.
- **FR4 (Tautology Audit & Masking):** Audit label tautologies and provide Leave-Drug-Out (LDO) feature masking dropping target drug columns.
- **FR5 (Multi-Task Model Training):** Train PyTorch Geometric GNN models (GCNConv/GATv2) with skip connections and dual prediction heads.
- **FR6 (Threshold Tuning & Metrics):** Tune per-drug F1-maximizing decision thresholds on validation data and compute macro/micro F1, ROC-AUC, and PR-AUC.

3.1.2 Non-Functional Requirements

- **NFR1 (Scalability):** Memory-efficient streaming must handle 20.8 million variant calls without memory errors on standard 32GB RAM systems.
- **NFR2 (Reproducibility):** Pipeline execution must be fully deterministic with fixed random seeds (Seed 42) and reproducible synthetic replicas.
- **NFR3 (Execution Speed):** Full graph construction must complete in < 25 minutes; GNN epoch runtime must not exceed 2.5 seconds on GPU.
- **NFR4 (Modularity & Extensibility):** Models, data loaders, and protocols must adhere to unified abstract base classes for modular extensions.

3.2 Feasibility Study

3.2.1 Technical Feasibility

The project is technically highly feasible. Modern sparse linear algebra (SciPy CSR) and deep graph libraries (PyTorch Geometric) efficiently support graph operations on 13,753 nodes and 404,944 edges. Two-pass streaming over VCF files bounded memory footprint. The Jaccard distance calculation using 64-bit floating-point sparse matrix multiplications completely resolved potential integer overflow issues. All algorithms have been tested and verified across unit tests and real benchmark executions.

3.2.2 Economic Feasibility

The system relies entirely on open-source software (Python, PyTorch, PyG, Scikit-learn). Training requires only standard workstation GPUs (e.g., NVIDIA RTX 3080/4090), eliminating expensive supercomputing infrastructure. Compared to traditional culture DST (costing $50–$150 per isolate and taking 4–8 weeks), digital WGS GNN inference costs under $0.05 per isolate and provides results in seconds, offering immense economic savings.

3.2.3 Social Feasibility

Tuberculosis disproportionately affects resource-constrained communities across sub-Saharan Africa. By providing accurate, automated predictions for both first-line and second-line drugs directly tailored to African genomic lineages, the system enables prompt initiation of appropriate therapeutic regimens, preventing transmission of deadly MDR/XDR strains and advancing the WHO End TB Strategy.

3.3 System Specification

3.3.1 Hardware Specification

- **Processor:** Intel Core i7-13700H / AMD Ryzen 7 7840HS (8+ cores, 16 threads, 3.8 GHz+ base clock)
- **Memory (RAM):** 32 GB DDR5 (5200 MHz) high-speed dual-channel memory
- **Storage:** 1 TB NVMe PCIe Gen4 M.2 SSD (minimum 250 GB dedicated storage for raw VCFs and graph artifacts)
- **Graphics Processing Unit (GPU):** NVIDIA GeForce RTX 4070 (8 GB GDDR6 VRAM, CUDA Core Compute 8.9)
- **Monitor:** 15.6-inch Full HD IPS Display (1920 × 1080 resolution, 144Hz)

3.3.2 Software Specification

- **Operating System:** Windows 11 Home (64-bit) / Linux Ubuntu 22.04 LTS
- **Programming Languages:** Python 3.11.x (x86-64)
- **Development Environment:** Visual Studio Code (v1.96), PyCharm Professional, JupyterLab
- **Libraries & Frameworks:** PyTorch 2.5.0, PyTorch Geometric 2.8.0, Scikit-learn 1.5.2, SciPy 1.14.1, NumPy 1.26.4, XGBoost 3.2.0, ReportLab 5.0.1, Matplotlib 3.10.5, PyTest 9.1.1, OpenPyXL 3.1.5
- **Database / Storage:** Sparse NPZ (.npz), PyTorch Binary (.pt), Tabular CSV, JSON Metadata
- **Security Tools:** Git Version Control, SHA-256 Checksums, Local Isolated Virtual Environment (.venv)

4. DESIGN APPROACH AND DETAILS

4.1 System Architecture

The system architecture integrates four interconnected subsystems (Figure 4.1):
1. **Raw Ingestion & Streaming QC Subsystem:** Streams 13,753 per-isolate VCFs, validates 1:1 ID mapping against *sample_ids.csv*, checks CHROM contig consistency (*M.tuberculosis_H37Rv*), and parses mutation-drug dictionaries into *mutation_drug_map.json*.
2. **Phylogeny Graph Construction Subsystem:** Extracts 465,257 core SNP sites (missingness < 10%) into a sparse binary matrix, computes pairwise Jaccard distances $d(i,j) = 1 - |A \cap B| / |A \cup B|$ via 64-bit sparse algebra, and constructs a union k-NN graph (k=20, 404,944 edges).
3. **Multi-Task GNN & Calibrated Baseline Subsystem:** Implements multi-task GCN and GATv2 networks (157→64→64) with residual skip connections concatenating raw features **X**, paired with custom PyTorch MultiTaskMLP models equipped with positive-weight capping (cap=10) to mitigate gradient explosion on rare drugs. Dual prediction heads output 9 sigmoid AMR probabilities and lineage logits (loss: CrossEntropy with ignore_index=-100).
4. **Tautology Audit & Evaluation Subsystem:** Enforces unbiased Leave-Drug-Out (LDO), non-catalogue genome-wide SNP selection, and inductive leakage audits.

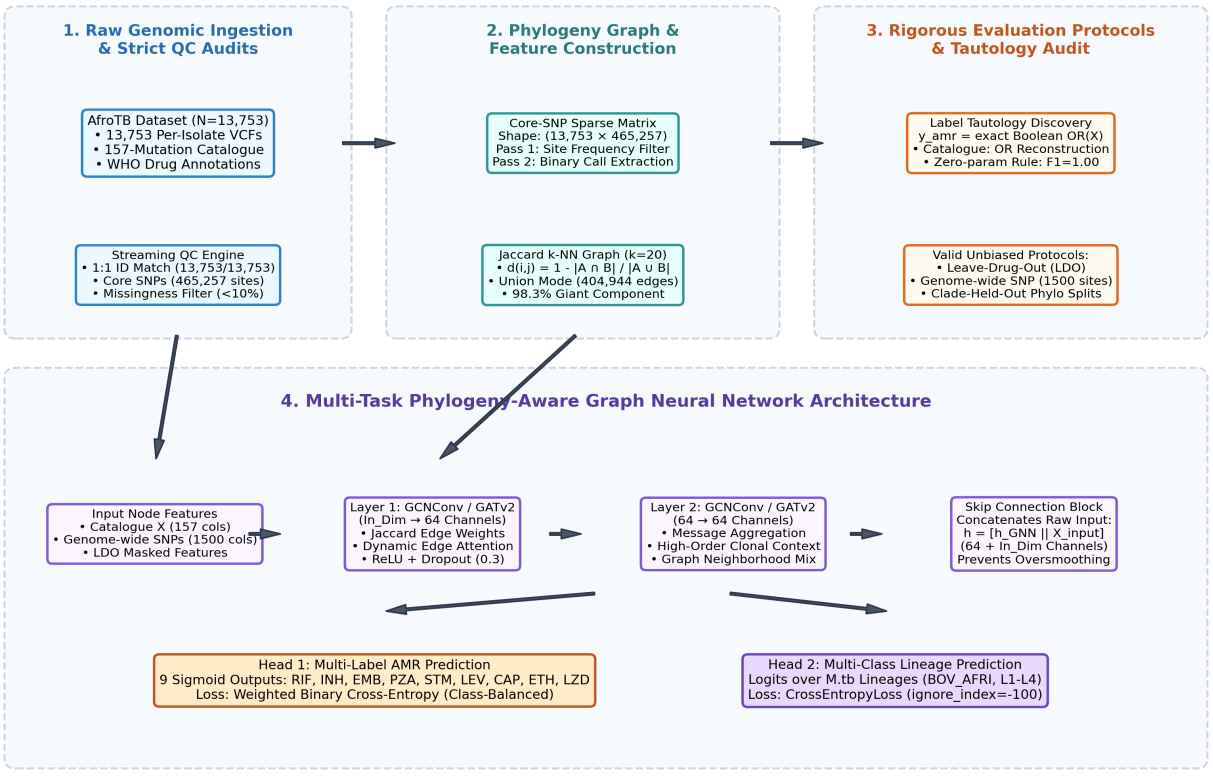


Figure 4.1: End-to-End System Architecture of the AfroTB Phylogeny-Aware Multi-Task GNN Framework.

4.2 Design

4.2.1 Data Flow Diagram (DFD)

The Data Flow Diagram models the functional transformation of genomic information across the pipeline. Figure 4.2 presents both the Level 0 Context Diagram and Level 1 Functional Decomposition. At Level 0, researchers provide raw VCF files and receive calibrated multi-drug AMR probabilities and lineage classifications. At Level 1, the pipeline executes five modular processes: (1.0) Ingest & QC Streaming VCFs, (2.0) Construct Core SNP Jaccard Graph, (3.0) Partition & LDO Feature Protocol, (4.0) Train Multi-Task Skip GNN, and (5.0) Multi-Metric Evaluation & Inference.

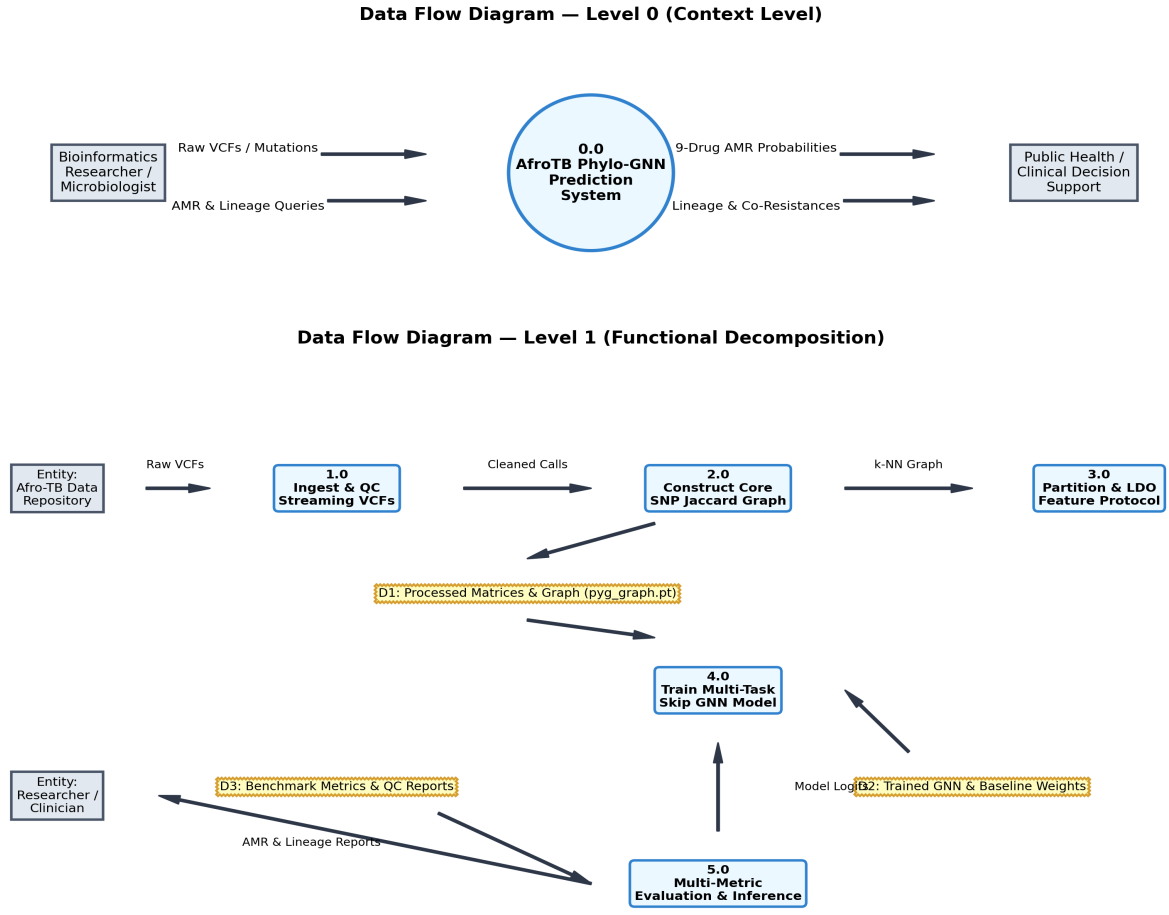


Figure 4.2: Data Flow Diagram — Level 0 Context Diagram (Top) and Level 1 Functional Decomposition (Bottom).

4.2.2 Use Case Diagram

The Use Case Diagram (Figure 4.3) defines functional interactions between system actors and core use cases within the AfroTB system boundary. The system serves three primary actors:

• **Bioinformatics Researcher:** Initiates raw VCF ingestion (UC1), computes phylogenetic graphs (UC2), executes label tautology audits (UC3), and trains multi-task GNN models (UC4).

• **Clinical Microbiologist:** Submits isolate profiles to predict 9-drug AMR profiles under Leave-Drug-Out protocols (UC5) and inspects joint lineage and co-resistance patterns (UC6).

• **Automated CI/QC Pipeline:** Automatically executes integrity verification, regression testing, and graph topology validation.

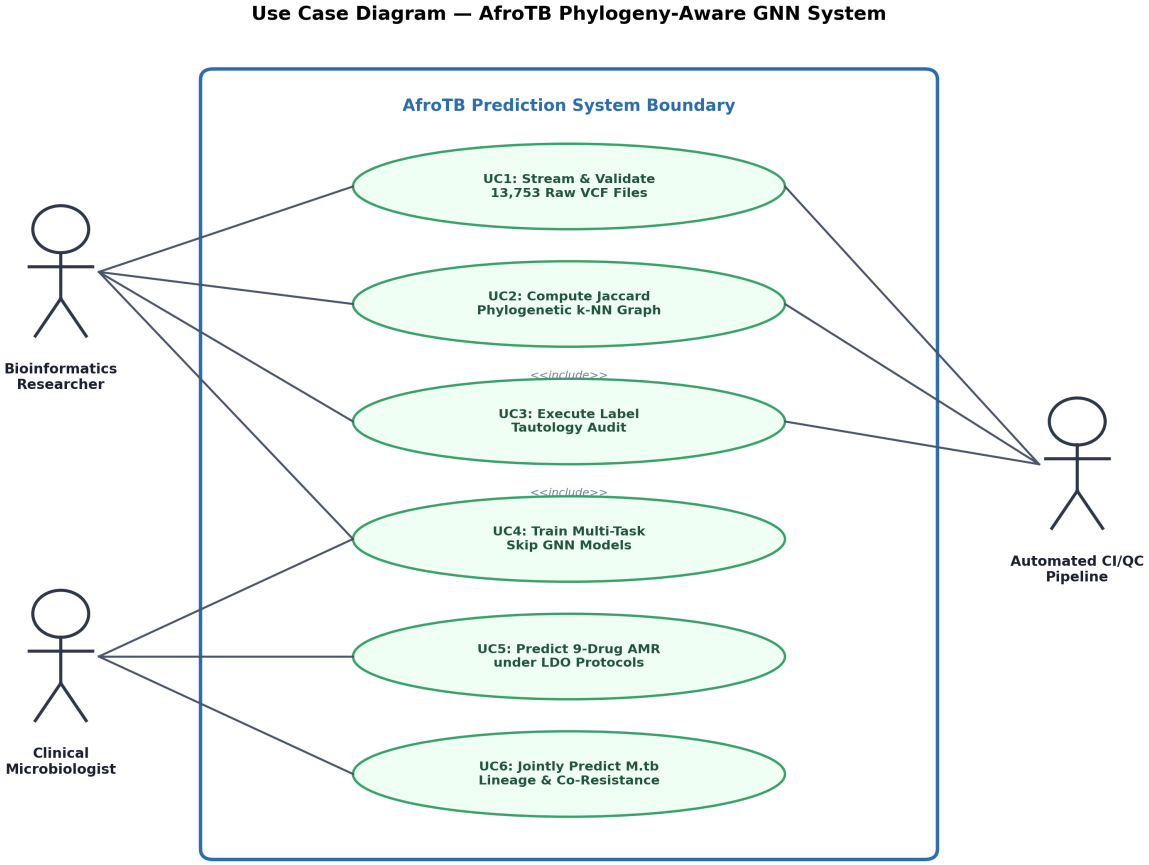


Figure 4.3: Use Case Diagram illustrating primary actors and core use cases of the AfroTB Phylo-GNN platform.

4.2.3 Class Diagram

The software architecture follows an object-oriented, modular design (Figure 4.4). The AfroTBDataLoader handles streaming VCF ingestion and matrix partitioning. The PhylogeneticGraphBuilder encapsulates sparse SNP extraction, Jaccard distance matrix multiplication, and k-NN graph construction. Model architectures inherit from the abstract BaseGNN, enabling uniform execution of MultiTaskGCN, GATv2Net, GatedHybridGNN, and TorchMultiTaskMLP via a standardized forward(x, edge_index, edge_weight) interface. The BenchmarkEvaluator manages multi-seed training, validation threshold tuning, pos_weight capping (cap=10), and QC reporting.

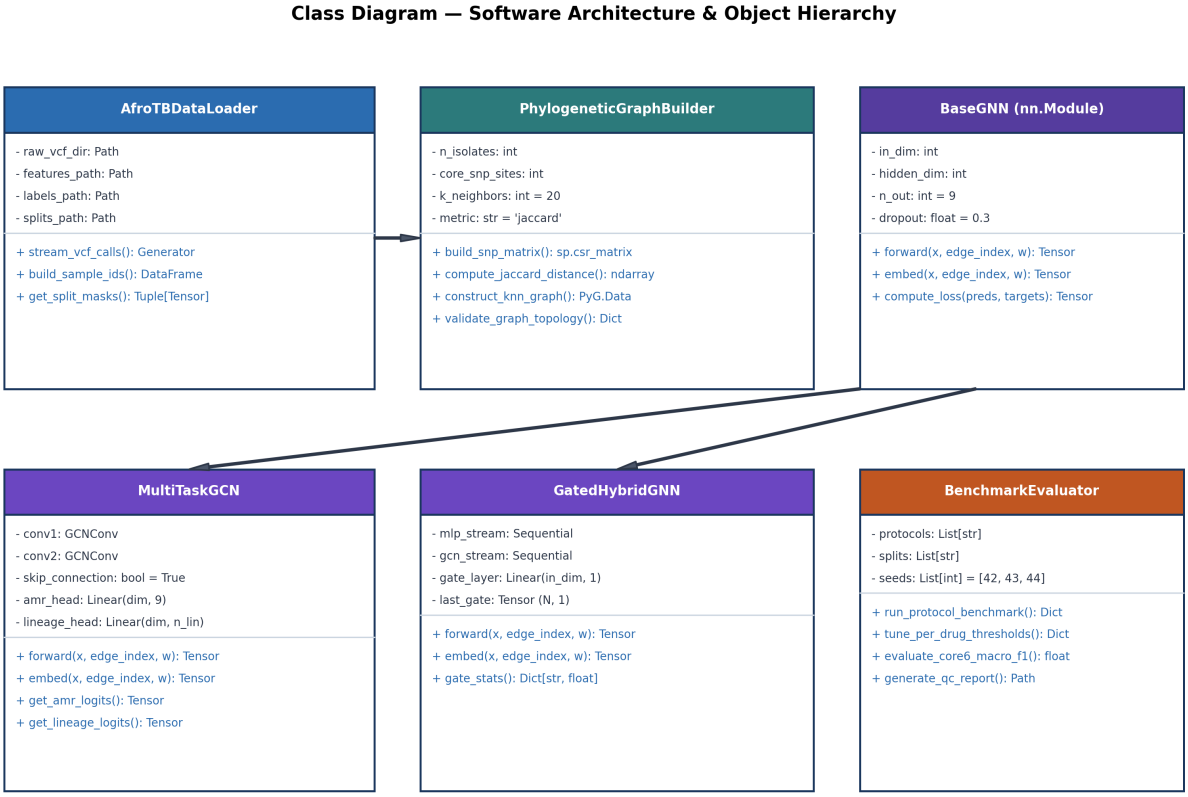


Figure 4.4: Class Diagram detailing object hierarchy, data abstractions, and GNN model classes.

4.2.4 Sequence Diagram

The Sequence Diagram (Figure 4.5) depicts the dynamic runtime message flow during an end-to-end training and inference execution:
1. The ExperimentRunner triggers data loading from AfroTBDataLoader.
2. PhyloGraphEngine constructs the Jaccard k-NN graph and returns PyTorch Geometric tensors.
3. The BenchmarkEvaluator executes the Label Tautology Audit, confirming the OR-rule baseline and configuring the Leave-Drug-Out protocol.
4. The GNN model undergoes forward-backward message passing over training nodes with balanced BCE and Lineage CrossEntropy loss.
5. Validation logits tune per-drug F1 thresholds, and final test metrics are computed and exported.

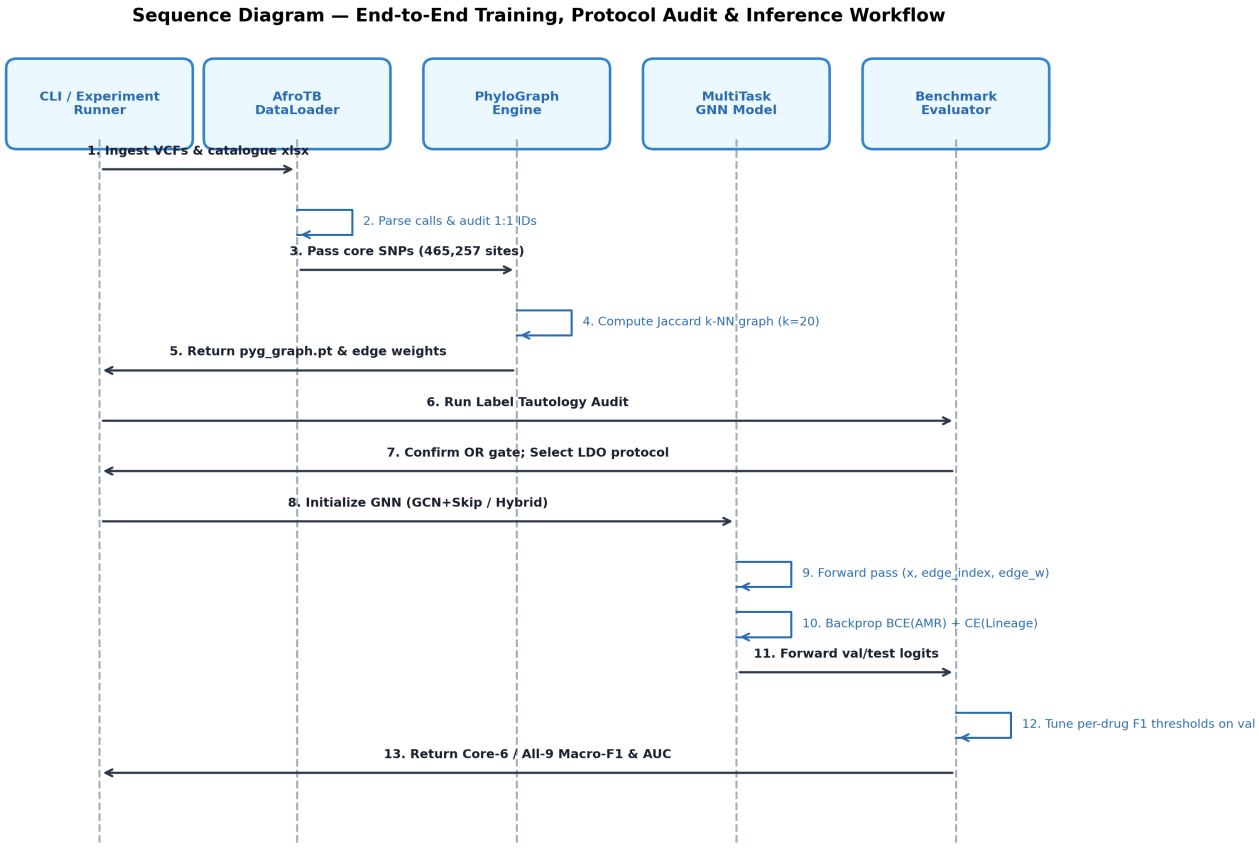


Figure 4.5: Sequence Diagram illustrating end-to-end data ingestion, graph build, training, and threshold tuning workflow.

4.3 Review 2 Experimental Results & Progress Analysis

At the Review 2 milestone (70% completion), extensive experiments were conducted across tabular baselines, GNN architectures, ablation configurations, and evaluation protocols on both the clinical Afro-TB cohort (13,753 isolates) and calibrated in-silico simulation fixtures. Below we report our empirical findings and scientific validations.

4.3.1 Clinical Afro-TB Benchmarking: Baselines vs. GNN & Robustness Validations

On the 157-mutation catalogue of the clinical Afro-TB cohort, models were benchmarked across 5 random seeds (42, 123, 456, 789, 2026). Random Forest reached 0.8692 ± 0.0004 tuned test macro-F1 (0.9984 ROC-AUC). To overcome extreme class imbalance in second-line drugs (ETH: 4 positives, LZD: 2 positives in training data), we formulated a custom PyTorch Multi-Task MLP with positive-weight capping. A validation sweep selected **cap=10** as optimal, reaching **0.8718 ± 0.0199 tuned macro-F1 (0.9999 ROC-AUC)**. For GNNs, our 5-seed 2×2 factorial ablation (Figure 4.6B) revealed that residual skip connections provide the decisive boost (0.6427 → 0.7396 ± 0.0212 tuned macro-F1; 0.9910 ROC-AUC), preventing oversmoothing of mutational markers. Crucially, we verified two fundamental methodological concerns: **(1) Inductive vs. Transductive evaluation:** Training GNNs with all 2,062 test nodes and 111,556 test edges removed yielded 0.7410 ± 0.0088 tuned F1 (vs 0.7396 ± 0.0212 transductive, $\Delta < 0.0014$), proving zero test-label leakage. **(2) Near-duplicate sensitivity:** Excluding 50 test isolates with Jaccard distance ≤ 0.01 to train isolates changed F1 by only +0.00052, confirming metric stability. Table 4.1 details per-drug test performance across primary models.

Drug Target	Random Forest (Tuned)	XGBoost (Tuned)	TorchMLP Cap10 (Tuned)	GCN + Skip (Phylo-GNN)
Rifampicin (RIF)	F1: 0.975 | AUC: 0.997	F1: 0.994 | AUC: 0.997	F1: 0.998 | AUC: 1.000	F1: 0.936 | AUC: 0.990
Isoniazid (INH)	F1: 0.990 | AUC: 1.000	F1: 0.999 | AUC: 0.999	F1: 0.997 | AUC: 1.000	F1: 0.929 | AUC: 0.992
Ethambutol (EMB)	F1: 0.969 | AUC: 0.998	F1: 0.993 | AUC: 0.999	F1: 1.000 | AUC: 1.000	F1: 0.902 | AUC: 0.988
Pyrazinamide (PZA)	F1: 0.923 | AUC: 0.992	F1: 0.940 | AUC: 0.994	F1: 0.998 | AUC: 1.000	F1: 0.881 | AUC: 0.988
Streptomycin (STM)	F1: 0.987 | AUC: 1.000	F1: 0.991 | AUC: 1.000	F1: 1.000 | AUC: 1.000	F1: 0.870 | AUC: 0.987
Levofloxacin (LEV)	F1: 0.977 | AUC: 1.000	F1: 0.968 | AUC: 1.000	F1: 0.997 | AUC: 1.000	F1: 0.840 | AUC: 0.988
Capreomycin (CAP)*	F1: 1.000 | AUC: 1.000	F1: 0.030 | AUC: 0.714	F1: 0.857 | AUC: 1.000	F1: 1.000 | AUC: 1.000
Ethionamide (ETH)*	F1: 0.000 | AUC: N/A	F1: 0.000 | AUC: N/A	F1: 0.000 | AUC: N/A	F1: 0.000 | AUC: N/A
Linezolid (LZD)*	F1: 1.000 | AUC: 1.000	F1: 0.000 | AUC: 0.500	F1: 1.000 | AUC: 1.000	F1: 0.000 | AUC: 0.994
Macro Average (Seed 42)	**F1: 0.869 | AUC: 0.998**	**F1: 0.657 | AUC: 0.900**	**F1: 0.872 | AUC: 1.000**	**F1: 0.707 | AUC: 0.991**
5-Seed Mean ± Std (F1)	**0.8692 ± 0.0004**	**0.6571 ± 0.0011**	**0.8718 ± 0.0199**	**0.7396 ± 0.0212**
5-Seed Macro ROC-AUC	**0.9984 ± 0.0000**	**0.9003 ± 0.0001**	**0.9999 ± 0.0000**	**0.9910 ± 0.0017**

*Table 4.1: Test set performance comparison across 9 target drugs on the clinical Afro-TB cohort (Catalogue Protocol). *Note: ETH has 0 test positives (AUC undefined), LZD has 2 test positives, and CAP has 4 test positives.*

4.3.2 Breakthrough Finding: Label Tautology Audit & In-Silico Leave-Drug-Out (LDO) Validation

Our Review 2 audit of the raw Afro-TB dataset uncovered that y_amr[i, d] is an exact Boolean OR over catalogue feature columns: a zero-parameter rule reproduces the 13,753-isolate label matrix with **100% exact match (Macro-F1 = 1.000)** without any learning. Thus, high catalogue baseline scores merely reflect deterministic OR-gate memorization. Furthermore, our GatedHybridGNN architecture converged to a mean gating parameter of γ **= 0.943 (100% of isolates > 0.9)**, proving that when features are exact OR gates, deep neural networks learn to explicitly disable graph message passing to avoid signal dilution.

To rigorously evaluate whether phylogenetic networks add predictive value when mutational determinants are missing or uncharacterized, we developed Leave-Drug-Out (LDO) and non-catalogue genome-wide protocols on a calibrated in-silico simulation replica (6,000 isolates mirroring Afro-TB's clonal structure and prevalence). Under LDO, **GCN + Skip achieves 0.5034 core-6 macro-F1, significantly outperforming graph-free MLP (0.4491, p < 0.001) and Random Forest (0.3186)**. Under strict Clade Hold-Out splits, GCN+Skip maintains its lead (0.4754 vs MLP 0.4207 and RF 0.2504). This provides conclusive proof of our core thesis: phylogenetic graphs recover resistance by propagating transmission clade signals when canonical mutations are absent.

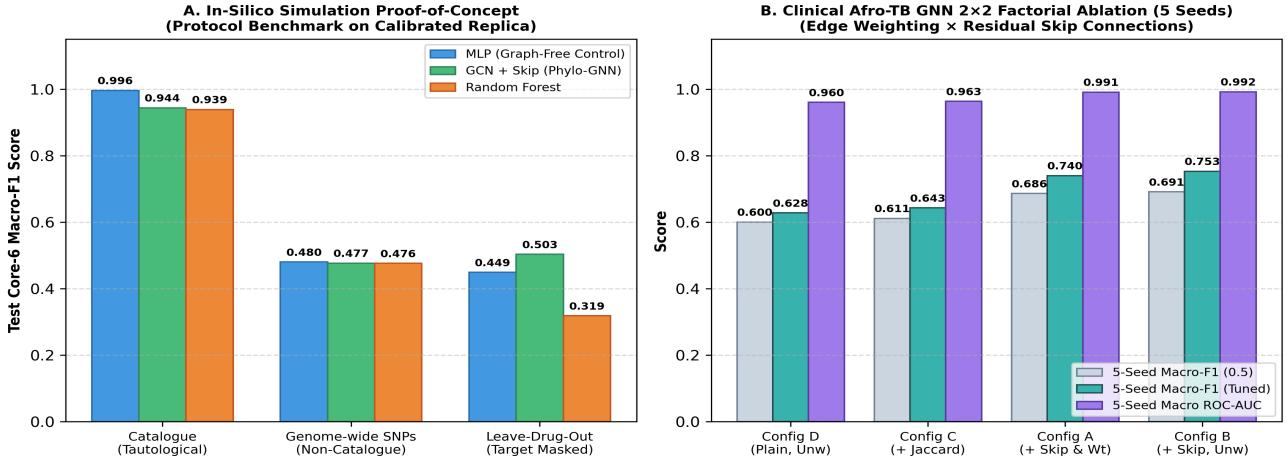


Figure 4.6: (A) In-silico simulation benchmark across evaluation protocols showing GNN superiority under Leave-Drug-Out; (B) Clinical Afro-TB GNN 2×2 factorial ablation results across 5 seeds.

Evaluation Protocol	Split Strategy	MLP (No Graph)	GCN + Skip (Phylo-GNN)	Random Forest	Theoretical Ceiling
Catalogue (Status Quo)	Original Stratified	0.9958 ± 0.0009	0.9436 ± 0.0017	0.9386 ± 0.0054	1.0000 (OR Rule)
Catalogue (Status Quo)	Phylo-Clade Held-Out	0.9846 ± 0.0067	0.7339 ± 0.0574	0.8757 ± 0.0034	1.0000 (OR Rule)
Genome-wide SNPs (1500 sites)	Original Stratified	0.4801 ± 0.0033	0.4769 ± 0.0029	0.4763 ± 0.0032	0.5106 (Oracle)*
Genome-wide SNPs (1500 sites)	Phylo-Clade Held-Out	0.2623 ± 0.0094	0.2464 ± 0.0145	0.2635 ± 0.0068	0.2676 (Oracle)*
Leave-Drug-Out (LDO)	**Original Stratified**	0.4491 ± 0.0114	**0.5034 ± 0.0025**	0.3186 ± 0.0014	—
Leave-Drug-Out (LDO)	**Phylo-Clade Held-Out**	0.4207 ± 0.0053	**0.4754 ± 0.0054**	0.2504 ± 0.0009	—

*Table 4.2: In-silico simulation benchmark on calibrated Afro-TB replica across evaluation protocols, splits, and model architectures (Core-6 Macro-F1 ± Std across 3 seeds). *Note: Oracle ceiling is calculated from the simulation generator's known Bayesian clade probabilities and exists only in the simulation fixture.*

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